

# Energy optimization analysis of a thermochemical exhaust gas recuperation system of a gas turbine unit



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## ABSTRACT

This article considers the scheme of a gas turbine unit (GTU) with a thermochemical exhaust heat recuperation system by using steam methane reforming. The main concept of thermochemical recuperation (TCR) is the transformation of exhaust gases heat into chemical energy of a new synthetic fuel that has higher calorimetric properties such as low-heating value. As an example, the gas turbine plants with turbines where the exhaust gas temperature exceeds 900 K are considered. To determine the optimum operating parameters of the thermochemical exhaust recuperation system, the influence of temperature, pressure, and inlet reaction mixture composition on the recuperation rate are determined. Based on thermodynamic analysis, the amount of exhaust heat that is transformed into chemical energy of the new synthetic fuel for various operating parameters is calculated. The thermodynamic analysis is performed by minimizing Gibbs energy via the programs IVTANT-HERMO and Aspen HYSYS. The results of the thermodynamic analysis are verified with the results obtained by the analytical calculation of other authors based on the law of mass action and the law of mass and energy conservation. As a result of the calculation, it was established that in the temperature range (900–1000) K the recuperation rate reaches a maximum value for the inlet reaction mixture composition of  $H_2O:CH_4 = 2$ ; in the temperature range of above 1200 K, at  $H_2O:CH_4 = 1$ . It is also established that when the pressure in the reaction space increases, the energy efficiency of the use of TCR is reduced; the optimum pressure is in the range of (5–10) bar. The maximum recuperation rate of the TCR system ( $R = 0.693$ ) is observed at  $T = 900$  K,  $\beta = 2$ ,  $p = 5$  bar.

## 1. Introduction

In the coming decades, fossil fuels, in particular natural gas, will remain as the primary energy sources for the world's heat-power engineering. According to the report of the International Energy Agency, the 21st century can be considered the "Golden Age of Gas" [1]. In addition, despite the latest advances in the use of renewable energy sources, the global demand share for natural gas will increase from 20% in 2011 to (26–28)% by 2030. [1,2].

Some of the most important consumers of natural gas are the gas turbine units (GTU) designed for the generation of electrical energy or mechanical work (shaft work of gas-pumping units) [3–5]. According to the second law of thermodynamics, one of the ways to improve the energy efficiency of the GTUs is to increase the temperature of the combustion gases at the gas turbine inlet [6,7]. Therefore, the modern versions of gas turbines have a temperature of combustion gases at the turbine inlet above 1300 K [8,9]. On the other hand, an increase in the temperature of the combustion products leads to an increase in the temperature of the exhaust flue gases. For example, for turbines SGT5-8000H and SGT6-8000H produced by Siemens, the temperature of the

exhaust flue gases is more than 900 K at an efficiency of about 40% [8,9].

To improve energy efficiency, GTUs are often combined with waste-heat recuperation systems. Usually, the waste-heat recuperation systems are presented as efficient solutions because the exhaust gas at the gas turbine outlet still has a moderately high temperature and contains much heat. The traditional approaches to improve the energy efficiency of GTUs are the integration with the steam power units, including the standard Rankine cycles (RCs) or the organic Rankine cycles (ORCs) [10–16]. Moreover, to improve energy efficiency in large GTUs, air and fuel are preheated by the exhaust gases before they enter the combustion chamber [17–19]. In GTUs, both additional RCs or ORCs and air/fuel preheaters are used as the exhaust gas recuperation system.

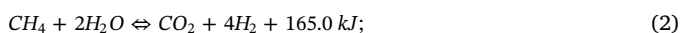
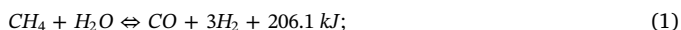
In recent years, the recuperation approach based on thermochemical transformation of the exhaust gases enthalpy to the chemical energy of a new synthetic fuel has found increasing interest among engineers and specialists [20–25]. This method of waste-heat recovery is called thermochemical recuperation (TCR) [26]. TCR can be a promising method to improve the energy efficiency of GTUs and other fuel-consuming equipment.

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TCR of exhaust gases heat takes place when chemical endothermic reactions are used, for example, steam methane reforming [20,23,27], combined steam-dry methane reforming [23–25,28], steam ethanol reforming [21], propane reforming [22], etc. The concept of increasing energy efficiency of fuel-consuming equipment by thermochemical waste-heat recuperation was discussed as early as 1972 [26]. Generally, various chemical endothermic reactions could be used for thermochemical waste-heat recuperation, but it is evident that the most advantageous is when reforming hydrocarbon gases are used, for example, steam methane reforming, which is described by the following chemical reactions:



The many thermochemical heat recuperation investigations have focused on gas turbine applications [20,29–31]. All of these papers highlighted the increase in lower heating value (LHV) of the new synthetic fuel compared to the initial (primary) fuel, as well as the increase of energy efficiency of this fuel-consuming equipment by using thermochemical waste-heat recuperation. Carapellucci and Milazzo compared the energy efficiency of GTUs with TCR and without TCR, and they concluded that the gas turbine with the steam methane reformer had better performance, for example, the coefficient of efficiency (52.4%) and power (288.4 MW) compared to the GTU performance without it (38.3% and 183 MW) [29]. Moreover, they suggested upgrading the original schematic diagram of TCR with one steam methane reformer to TCR with two reformers. By this upgrading, GTU with the thermochemical waste-heat recuperation system can reach the following performances:  $\eta = 53.8\%$ ;  $P = 292.9 \text{ MW}$ . This performance is fairly good compared with those of the GTU with Rankine cycles. This recuperation method is promising as it allows improved performances of GTU without a bottoming Rankine cycle.

Verkhivker and Kravchenko analyzed the energy efficiency of the combined cycle (gas turbine with steam cycle) with the TCR by steam methane reforming [20]. They concluded that the energy efficiency of the combined GTU with TCR can reach (80–90)%. Also, they reported that the exergy efficiency of the GTU with TCR has reached the level of (85–90)%. For TCR by steam methane reforming, the overall lower heating value of the new synthetic fuel can be 1.25 times higher than LHV of the original methane.

Han et al. developed a new power generation system based on the moderate conversion of coal and natural gas [32]. In this system, the heat of steam produced by the exhaust heat is used for reforming organic fuel (natural gas, coal). The authors calculated that with the same fuel input to power generation single systems, the new system based on the thermochemical principles can generate about 5.8% more electricity. The net efficiency of the new power generation system can achieve 48.6%, that is about 5% higher than that of the conventional combined gas cycles.

Moreover, Popov et al. showed that the energy efficiency of high-temperature plants with the thermochemical waste-heat recuperation was approximately 20 percentage points higher than that of the high-temperature plants without TCR [23]. In addition, Chakravarthy et al. published the results of thermodynamic analysis of an internal combustion engine (ICE) with thermochemical exhaust heat recuperation [25]. The authors concluded that the estimated second law efficiency can be increased for TCR with ethanol and isooctane reforming by 9 and 11%, respectively.

High-energy efficiency as the main advantage of TCR is highlighted for all the papers presented above. In addition, Knoche et al. [33] reported that the combustion irreversibility of the new synthetic fuel is less than that of the primary fuel. In summary, thermochemical exhaust heat recuperation is the promising method to increase the energy efficiency for the various fuel-consuming equipment, in particular, the gas

turbine units.

Furthermore, many publications on TCR for GTUs mainly focus on determining the energy efficiency of using this method of heat recovery. However, it is known that the temperature, the inlet gas composition of the initial reaction mixture, and the pressure in the reaction space have a significant effect on the TCR efficiency. This is due to the fact that the degree of conversion of the original fuel, for example, methane (degree of methane conversion), depends on these parameters. Many articles assume that the degree of conversion of the original fuel is 1, and the effect of temperature, pressure and feed gas composition are not accounted for.

The objectives of this paper are (1) to determine the operational conditions (temperature, feed gas composition and pressure) for optimal operation of the thermochemical exhaust gas recuperation system by steam methane reforming in the GTU; (2) to identify the advantages of the TCR in the GTUs.

## 2. GTU with thermochemical recuperation

The general idea of TCR applying to gas turbines is the increase in the lower heating value of the synthetic fuel (synfuel) compared to the original fuel converting the exhaust heat into chemical energy of a new synfuel. Fig. 1 illustrates the schematic diagram of the GTU with thermochemical exhaust heat recuperation by steam methane reforming. This diagram is based on two fundamental concepts: (1) the increase in the LHV of the synfuel compared to the original fuel; (2) the transformation of the exhaust heat into chemical energy of the new synfuel.

In this schematic diagram there are two subsystems. The first is the power generation subsystem with main components such as a compressor (C), a combustion chamber (CC), a gas turbine (GT) and a generator (G). The second is the thermochemical exhaust heat recuperation subsystem with main components such as a steam methane reformer (SMR) and a steam generator (SG). The exhaust gases from the GT are sequentially passing through the steam methane reformer and the steam generator. The steam methane reformer performs as a shell-and-tube heat exchanger. Heat is transferred through the wall. One side of the wall is filled with the exhaust gases, the other side is filled with the reaction mixture. The reaction mixture at the inlet consists of methane and steam. In the steam methane reformer, methane and steam are converted to the synthesis gas that is called the synthetic fuel or synfuel. The steam methane reforming absorbs a large amount of the high-temperature thermal energy that is extracted from the exhaust

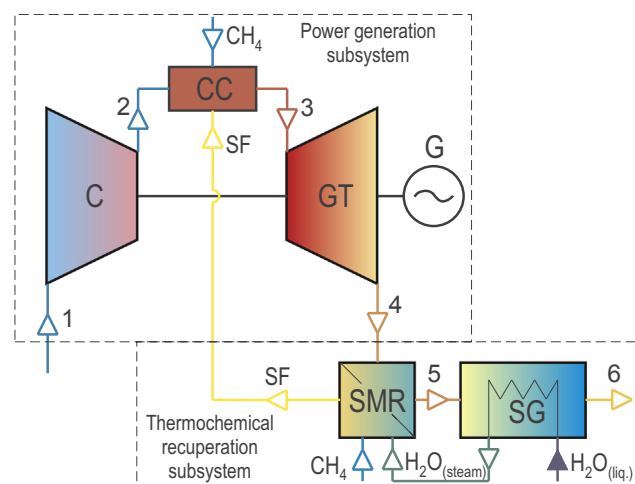


Fig. 1. The schematic diagram of the GTU with thermochemical exhaust heat recuperation by steam methane reforming: C – compressor; CC – combustion chamber; GT – gas turbine; G – generator; SMR – steam methane reformer; SG – steam generator; SF – synthetic fuel.

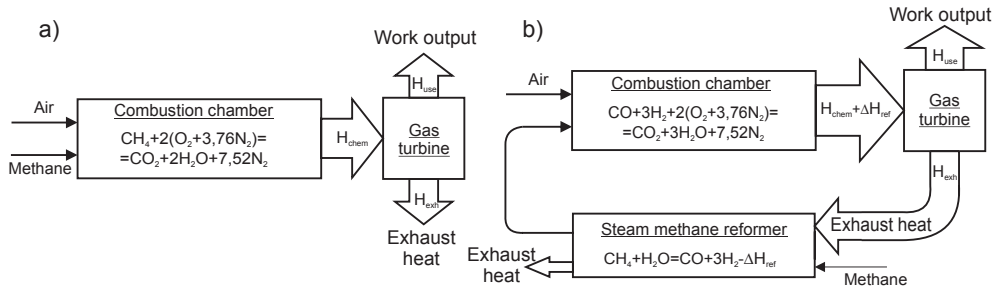


Fig. 2. The methods of heat extraction from the original fuel for the conventional GTU (a) and for GTU with TCR (b).

gases. The synfuel at the SMR outlet consists of the different gases ( $H_2$ ,  $CO$ ,  $CH_4$ ,  $CO_2$  and others). The synfuel from the steam methane reformer is fed into the CC (combustion chamber) of the GT (gas turbine) [34,35]. Steam for methane reforming is produced in the steam generator (SG) through the use of the residual exhaust heat.

In conventional GTUs, natural gas is burned directly in the combustion chamber. For example, when burning  $1\text{ m}^3$  of methane, 35.82 kJ is generated [36–39]. In the GTU with TCR, natural gas is initially converted to synthesis gas in the steam methane reformer. For example, when burning synthesis gas produced by full steam reforming of  $1\text{ m}^3$ , 45.03 kJ is generated. In summary, we can say that in the GTU with TCR the heat generation from the original fuel is divided into two stages. The first stage is the increasing of the lower heating value of the original fuel by the transformation of exhaust heat ( $H_{exh}$ ) into the chemical energy of synfuel. The second stage is the combustion of synfuel that has higher LHV than that of the original fuel. The methods of heat extraction from the original fuel for the conventional GTU (a) and for the GTU with TCR (b) are presented in Fig. 2.

The main criterion for the efficiency of the heat recuperation system is the recuperation rate. The recuperation rate shows how much heat of the exhaust gases is recovered in the GTU cycle. Obviously, the efficiency of using the thermochemical exhaust heat recuperation system is determined by the degree of conversion of the original fuel (degree of methane conversion). On the other hand, the degree of methane conversion depends on operating temperature, pressure and methane/steam feed composition. Therefore, the recuperation rate depends on operating temperature, pressure and methane/steam feed composition. In the next section, the algorithm for calculating the recuperation rate is given.

### 3. Methodology

#### 3.1. Recuperation rate

According to definition, the recuperation rate can be determined as:

$$R = \frac{\Delta H_{rec}}{H_{exh}} = \frac{H_{exh} - H_{out}}{H_{exh}} \quad (4)$$

where  $\Delta H_{rec}$  – recuperated heat in the thermochemical recuperation system, kJ;  $H_{exh}$  – exhaust heat after the gas turbine (pos.4 in Fig. 1), kJ;  $H_{out}$  – enthalpy of the waste flue gases (pos.6 in Fig. 1), kJ.

The problem of determining the recuperation rate in the thermochemical exhaust heat recuperation system is a conjugate problem with the thermodynamic analysis of the steam methane reforming process. This is due to the fact that the synfuel composition, the degree of methane conversion, the enthalpy of the conversion reaction, etc. depend on the operational conditions such as temperature, pressure, and composition of the inlet reaction mixture. The interconnection of these parameters allows the establishing of a thermodynamic analysis [22,40].

The exhaust heat after the gas turbine and enthalpy of the waste flue gases after the TCR system are expressed by:

$$H_{exh} = G_{exh} \cdot c_{p_{exh}} \cdot T_{exh}; H_{out} = G_{out} \cdot c_{p_{out}} \cdot T_{out}, \quad (5)$$

where  $G_{exh}$ ,  $G_{out}$  – the mass flow of the exhaust gases and waste flue gases, respectively, kg/s;  $T_{exh}$ ,  $T_{out}$  – the temperature of the exhaust gases after gas turbine and waste flue gases after the TCR system, respectively, K;  $c_{p_{exh}}$ ,  $c_{p_{out}}$  – the specific heat of the exhaust gas and waste flue gas, respectively, kJ/(kg·K).

In Eq. (5)  $G_{exh} = G_{out}$ , because it is assumed that the air inleakage for the thermochemical recuperation system is equal to zero.

According to Fig. 1, in the TCR system, the exhaust heat is used to carry out the reaction of steam methane reforming and to heat the reaction components. However, synthetic fuel consists of combustible components (hydrogen, carbon monoxide, unreacted methane) and non-combustible components (unreacted steam, carbon dioxide formed by water–gas shift reaction (3)). Therefore, the enthalpy of incombustible components cannot be considered as a recuperated heat because in this case, they are the ballast components. In summary, the recuperated exhaust heat in the TCR system can be determined by the following expression:

$$\Delta H_{rec} = H_{SMR} + H_{SF} - H_{H_2O} - H_{CO_2} - H_{fw} - H_{CH_4}, \quad (6)$$

where  $H_{SMR}$  is the enthalpy of steam methane reforming reaction, kJ;  $H_{SF}$  is the enthalpy of the synthetic fuel after steam methane reformer (pos.SF in Fig. 1), kJ;  $H_{H_2O}$ ,  $H_{CO_2}$  are the enthalpy of  $H_2O$  and  $CO_2$  that are contained in the synfuel, kJ;  $H_{fw}$  is the enthalpy of feed water before the steam generator (pos.  $H_2O_{(liq)}$  in Fig. 1, kJ;  $H_{CH_4}$  is the enthalpy of methane at the steam methane reformer, kJ.

Obviously, the terms of expression (6) such as  $H_{SMR}$ ,  $H_{SF}$ ,  $H_{H_2O}$ ,  $H_{CO_2}$  is depending on the operation conditions such as temperature, pressure, composition of the inlet reaction mixture. To determine  $H_{SMR}$ ,  $H_{SF}$ ,  $H_{H_2O}$ ,  $H_{CO_2}$  it is important to determine the composition of the synfuel and the enthalpy of the steam methane reforming reaction. This can be done with the help of a thermodynamic analysis of the steam methane reforming process.

Therefore, the algorithm of calculation of the recuperation rate can be developed as shown in Fig. 3:

#### 3.2. Thermodynamic analysis

The main goal of thermodynamic analysis is the calculation of synthetic fuel compositions at various operating conditions such as temperature, pressure and inlet gas composition. The thermodynamic analysis makes it possible to determine the limiting (equilibrium) synfuel compositions by using the method of Gibbs free energy minimization (GFEM). The basic concept of GFEM is built on the principle that the thermochemical system is thermodynamically favorable when its total Gibbs energy is at its minimum value and its differential is zero for each pressure and temperature [21,22,40,41]. The basic concept of GFEM can be written as:

$$(dG^t)_{T,p} = 0 \quad (7)$$

The algorithm of the determination of total Gibbs energy was discussed in author's previous work [22]. The resulting equation for

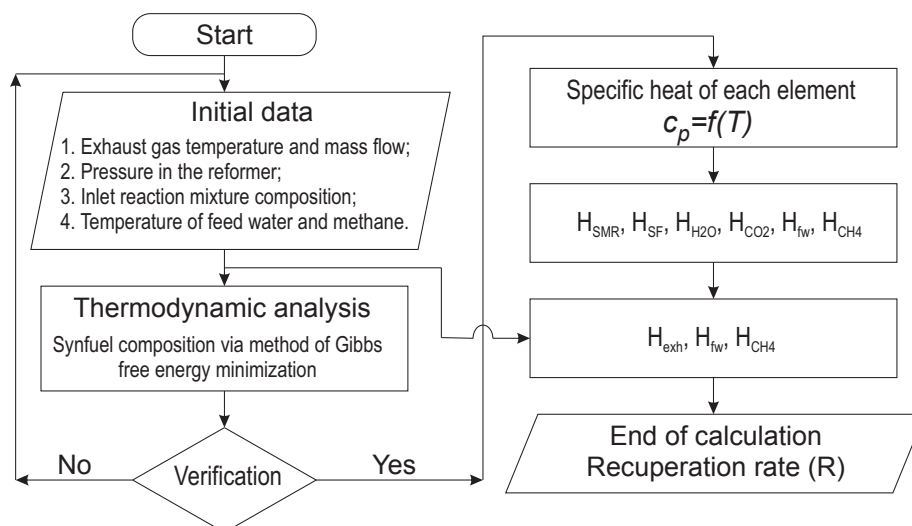


Fig. 3. The algorithm of calculation of the recuperation rate.

determination of total Gibbs energy is defined as follows:

$$G^t = \sum n_i \Delta G_{fi}^0 = \sum n_i RT \ln \left( \frac{f_i}{f_i^0} \right) \quad (8)$$

where  $n_i$  – number of moles of element  $i$ ;  $G_{fi}^0$  – standard Gibbs energy of formation of element  $i$ ;  $R$  – molar gas constant;  $T$  – temperature;  $f_i$  – fugacity of pure element  $i$ ;  $f_i^0$  is fugacity of pure element  $i$  at the standard state.

The thermodynamic analysis by using the GFEM technique can be done by employing various special programs. In this paper, the thermodynamic equilibrium analysis is performed by using two programs IVTANTHERMO and Aspen HYSYS. The IVTANTHERMO is a process simulator for thermodynamic modelling of chemical systems [42–44]. The IVTANTHERMO software contains an extensive database of thermodynamic properties of chemical components, algorithms for the database computing, and a programs, which allow the determination of equilibrium composition and thermodynamic properties of the chemical system to be examined. The results were checked by Aspen-HYSYS version 8.4, which is a powerful program for modelling unsteady-state and steady-state chemical processes for different systems [45–48]. Also, both programs calculate the distribution of reaction products by minimizing the Gibbs energy of each species in the reaction system regardless of the reaction network. The calculation model is presented in Fig. 4. Both programs use for calculation the fundamental principles of the law of mass action and the law of mass and energy conservation.

The major components of steam methane reforming reaction are

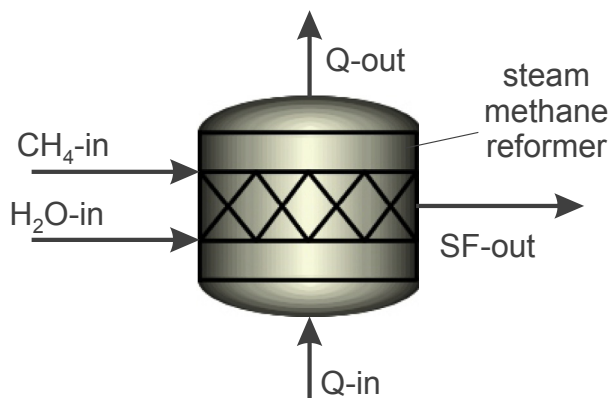


Fig. 4. The calculation model of the steam methane reforming process.

hydrogen ( $H_2$ ), carbon monoxide ( $CO$ ), methane ( $CH_4$ ), carbon dioxide ( $CO_2$ ) based on experimental results. The steam methane reforming reactions (1) and (2) and water gas-shift reaction are also accompanied by the different chemical reactions. However, according to the thermodynamic analysis in [49,50], only the mentioned three reactions play a significant role in steam methane reforming process while the rates of the other reactions are negligible.

In addition, the following assumptions have been made:

1. Steady state operation.
2. Isothermal conditions prevail.
3. Negligible pressure drop.
4. Plug flow in the virtual reactor.
5. No interphase and intraparticle mass transfer limitations.

The fifth assumption was confirmed by the experiments and the theoretical calculation [51]. The second assumption is the key factor for the thermodynamic analysis. The third assumption was confirmed by the experimental investigations of the pressure drop in the packed bed reformer [52,53]. Therefore, the neglect of pressure drop through the catalyst bed is reasonable. The fourth assumption can be accepted under the conditions of the high temperatures and the moderate low pressures used.

## 4. Results and discussion

### 4.1. Verification

To validate obtained results in the present thermodynamic analysis, the synthetic fuel gas compositions were compared with another thermodynamic study of Verkhivker and Kravchenko [20]. The thermodynamic analysis by minimizing Gibbs energy can be used to calculate the maximum gas composition that may be reached in the steam methane reforming system at a constant temperature and pressure at equilibrium condition. There are articles with the results of experimental investigation at the different operational conditions [49,54–56]; however, the difference in the conditions of the experiments makes it impossible to adequately compare these results with the results of thermodynamic analysis. In addition, in Ref. [57], the results of investigations of steam methane reforming for a wide range of operational conditions are reported. The authors conclude that almost equilibrium gas composition can be achieved at the high residence time.

Thermodynamic calculations were performed for different temperatures, pressures, and inlet reaction mixture compositions. A

**Table 1**

The comparison of the synfuel gas composition, % (mol/mol).

| Temperature | Pressure | Source          | Gas composition, % |       |                  |                 |                 |
|-------------|----------|-----------------|--------------------|-------|------------------|-----------------|-----------------|
|             |          |                 | H <sub>2</sub>     | CO    | H <sub>2</sub> O | CO <sub>2</sub> | CH <sub>4</sub> |
| 1000 K      | 1 bar    | IVTAN           | 63.01              | 14.32 | 16.32            | 5.23            | 1.12            |
|             |          | ASPEN           | 63.40              | 14.38 | 16.12            | 5.08            | 1.02            |
|             |          | Verkhivker [20] | 63.42              | 14.43 | 16.11            | 5.12            | 0.92            |
| 1000 K      | 5 bar    | IVTAN           | 60.66              | 14.82 | 17.32            | 5.55            | 1.65            |
|             |          | ASPEN           | 61.60              | 14.98 | 17.92            | 4.08            | 1.42            |
|             |          | Verkhivker [20] | 61.44              | 14.93 | 17.97            | 4.18            | 1.48            |
| 1100 K      | 1 bar    | IVTAN           | 63.88              | 16.08 | 16.02            | 3.93            | 0.09            |
|             |          | ASPEN           | 63.65              | 16.14 | 16.15            | 3.98            | 0.08            |
|             |          | Verkhivker [20] | 63.76              | 16.11 | 16.11            | 3.94            | 0.08            |
| 1100 K      | 5 bar    | IVTAN           | 61.28              | 15.02 | 17.87            | 4.24            | 1.59            |
|             |          | ASPEN           | 61.57              | 14.99 | 17.88            | 4.11            | 1.45            |
|             |          | Verkhivker [20] | 61.45              | 14.93 | 17.96            | 4.18            | 1.48            |

comparison of the results of the synfuel composition is shown in Table 1. The upper results are the mole fraction of each component that was calculated with the help of IVTANTHERMO, the middle results are the mole fraction that was obtained with Aspen-HYSYS, the lower results are the mole fraction that was taken from Verkhivker's work [20], in which an analytical calculation of the steam methane reforming was done. For all the results obtained, good convergence is observed.

The comparison of the thermodynamic analysis results with the results of other authors [58,59] showed a good correlation for the wide range of the operational conditions.

Carbon formation is another factor that needs to be considered, especially at high temperatures. However, there is sufficient information in literature that coke formation can be controlled by improving the coking resistance of the catalysts as well as by regulating the operating conditions such as steam-to-methane ratio [60]. In the present investigation, steam-to-methane ratio for all inlet gas compositions is either equal to or greater than unity, which indicates that there is enough steam to oxidize and prevent formation of solid carbon [61]. Hence, carbon formation was omitted from equilibrium expressions.

Moreover, the coke deposition can be eliminated by choosing the right operational conditions and situation. There are new catalysts with complicated mechanisms that can minimize coke deposition effectively [62]. According Ref. [58], the maximum coke formation for steam methane reforming can be observed at  $T = 1103$  K and  $\beta = 0.55$ . But the optimal operation conditions for the TCR system show a significant difference.

#### 4.2. Reaction enthalpy

According to the first law of thermodynamics, the enthalpy of flue gases transformed into the chemical energy of a synthetic fuel is determined by the enthalpy of the steam methane reforming reaction (Eq. (1)). The overall contribution to the material and energy balance of water–gas shift reaction Eq. (3) is insignificant and may not be taken into account in thermodynamic analysis [49]. This disregard is due to the fact that in the temperature range of more than 600 K, the change in the free Gibbs energy for reaction (Eq. (1)) is substantially greater than the change in the Gibbs energy for reaction (Eq. (3)).

Fig. 5 shows the enthalpy of the steam methane reforming reaction in the temperature range (600–1350) K at a pressure of 10 bar for various compositions of the initial reaction mixture of  $\text{H}_2\text{O}:\text{CH}_4 = \beta = 1 \dots 6$ .

As can be seen from Fig. 5, the maximum value of the reaction enthalpy of about  $9200 \text{ kJ/m}^3$  corresponds to a temperature range above 1200 K for all  $\text{H}_2\text{O}:\text{CH}_4 = \beta$  ratios. At the same time, for the temperature of about 900 K, which is corresponding to the temperature

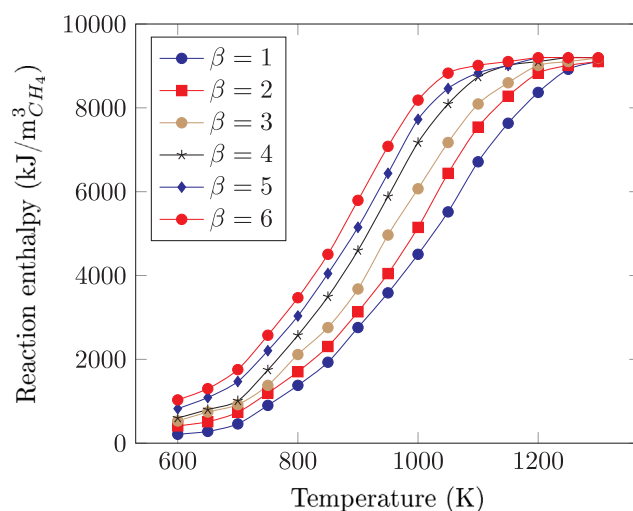


Fig. 5. The enthalpy of the SMR reaction as a function of temperature for various compositions of the initial reaction mixture  $\beta$  at  $p = 10$  bar.

of the exhaust gases of turbines SGT5-8000H and SGT6-8000H (produced by Siemens), the reaction enthalpy strongly depends on the amount of steam in the initial reaction mixture. For example, for the stoichiometric amount of the reaction components ( $\beta = 1$ )  $H_{\text{SMR}} = 2314 \text{ kJ/m}^3$ , then for  $\beta = 5$  the reaction enthalpy is more than twice higher.

Obviously, the energy efficiency of using the thermochemical heat recovery of exhaust gases depends on the enthalpy of the steam methane reforming reaction. If the SMR reaction does not take place in the reformer, then the energy efficiency of the use of thermochemical recovery is zero or even has a negative value, since for steam production, exhaust heat is used. On the other hand, increasing the reaction enthalpy in the temperature range to (1300–1500) K by adding of steam leads to an increase of the volume of non-combustible components in the synthetic fuel, in particular steam. In this case, excess steam, although a part of the synthetic fuel, does not positively participate in the combustion process from the energy point of view. Therefore, excess steam is ballast. In addition, for the production of additional steam ( $\beta > 1$ ) in the thermochemical recovery system (according to the scheme shown in Fig. 1), additional heat of the exhaust flue gases is required. To determine the optimal operation parameters such as temperature, inlet gas composition, and pressure, the recuperation rate (recovery rate) is calculated and analyzed

#### 4.3. Recuperation rate

The recuperation rate for various inlet reaction mixture compositions and temperatures can be defined as:

$$R = \frac{\Delta H_{\text{rec}}}{H_{\text{exh}}} \quad (9)$$

For Eq. (9), the  $\Delta H_{\text{rec}}$  and  $H_{\text{exh}}$  are recuperated exhaust heat and exhaust heat, which are determined by Eqs. (6) and (5), respectively.

For further analysis of the scheme presented in Fig. 1, the following parameters are set:

- the methane temperature at the reformer inlet is 293 K;
- the feedwater temperature for the steam generator is 343 K.

Fig. 6 shows the dependence of the recuperation rate of the exhaust heat from the gas turbine plant versus the temperature for various feed gas compositions. According to the Le Chatelier-Brown principle, the steam methane reforming process is affected not only by the feed gas composition and the temperature but also by pressure. This is due to the

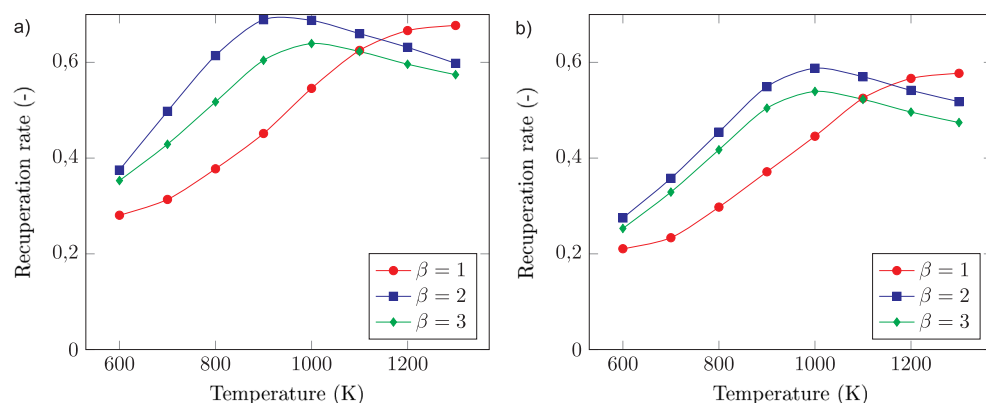


Fig. 6. The recuperation rate as a function of temperature for various compositions of the initial reaction mixture  $\beta$  at  $p = 5$  bar (a) and  $p = 10$  bar (b).

fact that the SMR reaction proceeds with an increase in the volume of the reaction products. This fact is also confirmed by the results given in Table 1. Therefore, Fig. 6(a) and (b) shows the recuperation rate at  $p = 5$  bar (a), and  $p = 10$  bar (b), respectively.

As can be seen from Fig. 6, with increasing pressure in the reaction space, the recuperation rate decreases. This is due to the fact that the methane conversion degree (relation of reacted methane (mol) in reaction (1) to inlet methane (mol)), which is basically determining the amount of transformed exhaust heat into chemical energy, depends on the pressure in the reaction space of the reformer. In the temperature range below 1100 K, the pressure in the reaction space of the reformer has a significant effect on methane conversion rate. According to Fig. 7, the methane conversion degree at pressure 5 bar is more than five times higher than the conversion degree at 10 bar for  $T = 1000$  K.

Fig. 6 shows that the maximum recuperation rate is observed in the temperature range (900–1100) K for all investigated cases. In addition, the maximum recuperation rate  $R = 0.693$  for a pressure of  $p = 5$  bar is observed at  $T = 900$  K and  $\beta = 2$ . For a pressure  $p = 10$  bar, the maximum recuperation rate is observed at  $T = 1000$  K and  $\beta = 2$ .

Analyzing the dependences in Fig. 6, the following facts can be established. When the  $\text{CH}_4:\text{H}_2\text{O}$  is higher than the stoichiometric  $\beta > 1$ , a sharp increase in the recuperation rate is observed. It can be due to a steep increase in the methane conversion degree in the temperature range up to 900 K (according to Fig. 7 and Table 1). In this temperature range, the main contribution to recovered heat is made by methane preheating, because at  $T < 900$  K the enthalpy of the steam methane reforming reaction is minimal. In other words, the exhaust heat recuperation is carried out basically according to the traditional scheme due to the heating of the combustion component, in the presented case it is methane.

The maximum recuperation rate for both pressures is observed in the temperature range of (900–1100) K. This is due to the fact that in

this temperature range there is an intensive steam methane reforming reaction. In this temperature range, the sufficiently high methane conversion degree is achieved. The main contribution to exhaust heat recuperation is the transformation of exhaust heat into the chemical energy of the new synthetic fuel by the steam methane reforming process.

In the temperature range above 1100 K, there is a gradual decrease in the recuperation rate due to the presence in the synthetic fuel of a large amount of ballast (non-combustible) components. These non-combustible components at high temperature have a noticeable enthalpy, while the reaction enthalpy almost reaches its maximum. In other words, the enthalpy of ballast substances in synthetic fuel increases faster than the enthalpy of the steam methane reforming reaction.

In addition, for a stoichiometric amount of the reaction components of the feed stock ( $\beta = 1$ ), a constant increase in the recuperation rate is observed. In the temperature region above 1100 K, the increase of the recuperation rate is slowed due to the achievement of the maximum reaction enthalpy. In the temperature range the increase of recuperation rate is observed mainly due to preheating of synthetic fuel in reformer. The recuperation rate is not decreasing in this temperature range because the minimum amount of ballast substances in the reforming products are observed in comparison to synfuel gas composition at  $\beta > 1$ . But in the temperature range above 1200 K the equilibrium of the water–gas shift reaction (Eq. (3)) is shifted to the right. As a result, a significant amount of carbon dioxide is observed in the conversion products, while the steam amount is minimum.

In summary, as can be seen from Fig. 6 the maximum energy efficiency of the TCR system is in the temperature range of (900–1000) K for  $\beta > 1$ ; in the temperature range of above 1200 K for  $\beta = 1$ , for all investigated pressures. To better understand the effect of  $\beta$  ratio on recuperation rate, Fig. 8 shows the recuperation rate of the exhaust heat versus the  $\beta$  ratio for the various temperatures.

Fig. 8 depicts the effect of  $\beta$  ratio on recuperation rate at  $p = 5$  bar. It can be seen that the maximum recuperation rate is achieved at  $\beta = 2$  in the temperature range of below 1200 K. For all investigated cases, the highest recuperation rate is observed for  $T = 900$  K at  $\beta = 2$ . This temperature corresponds to the temperature of the exhaust flue gases, for example, of gas turbines SGT5-8000H and SGT6-8000H (produced by Siemens). According to the material balance, for the maximum effectiveness of implementation of TCR by steam methane reforming at temperature of 900 K, approximately 1.603 kg of feed water must be supplied to the TCR system of the GTUs for each  $1 \text{ m}^3$  of the primary natural gas.

## 5. Conclusion

In this paper, the possibility in principle to use the thermochemical exhaust heat recuperation system by steam methane reforming in the gas turbine units is shown. The TCR uses the exhaust heat to supply the

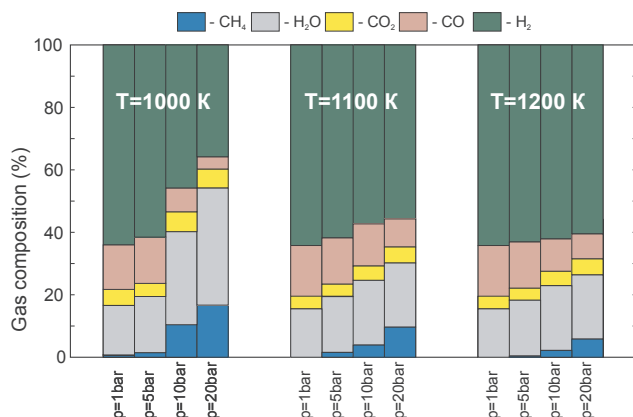


Fig. 7. The effect of pressure on reformat gas compositions.

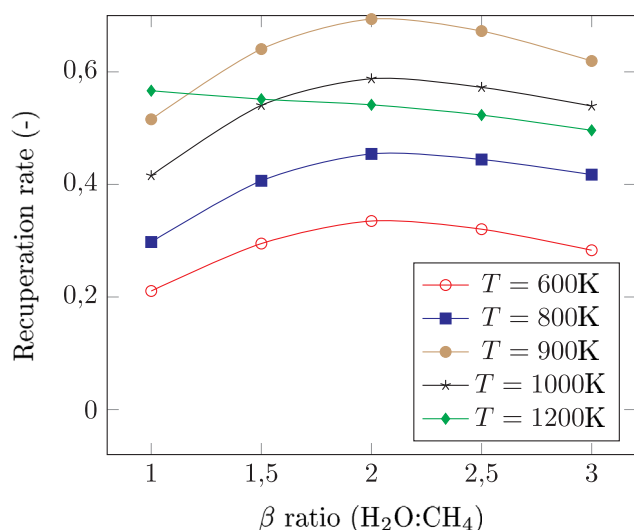


Fig. 8. The effect of  $\beta$  ratio on the recuperation rate at  $p = 5$  bar.

energy required for the steam methane reforming. The determination of the energy efficiency of the use of TCR is based on the first law of thermodynamics (energy conservation law) and thermodynamic analysis. The main conclusions obtained as the results of the research are:

- the operational parameters such as temperature, composition of the inlet reaction mixture and pressure exert a significant influence on the energy efficiency of using the thermochemical exhaust heat recuperation system;
- the steam addition to the inlet reaction mixture increases the enthalpy of the steam methane reforming reaction and, as a result, increases the value of exhaust gases enthalpy that is transformed into the chemical energy of the new synthetic fuel; at the same time, the addition of steam to the feed stock leads to an increase in the amount of non-combustible components in synthetic fuel;
- in the temperature range (900–1100) K the recuperation rate reaches the maximum value at a ratio of  $\text{H}_2\text{O}:\text{CH}_4 \approx 2$ ; in the temperature range of above 1200 K - at  $\text{H}_2\text{O}:\text{CH}_4 = 1$ ;
- the maximum recuperation rate ( $R = 0.693$ ) is observed at  $T = 900$  K,  $\beta = 2$ ,  $p = 5$  bar;
- with increasing pressure in the reaction space, the energy efficiency of the use of the thermochemical recuperation system is reduced; therefore the optimum pressure is pressure in the steam generator, for example, the pressure range of (5–10) bar.

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